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Gender differences in peer reviews of grant applications: A substantive-methodological synergy in support of the null hypothesis model☆

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ABSTRACT

Peer review serves a gatekeeper role, the final arbiter of what is valued in academia, but is widely criticized in terms of potential biases—particularly in relation to gender. In this substantive-methodological synergy, we demonstrate methodological and multilevel statistical approaches to testing a null hypothesis model in relation to the effect of researcher gender on peer reviews of grant proposals, based on 10,023 reviews by 6233 external assessors of 2331 proposals from social science, humanities, and science disciplines. Utilizing multilevel cross-classified models, we show that support for the null hypothesis model positing researcher gender has no significant effect on proposal outcomes. Furthermore, these non-effects of gender generalize over assessor gender (contrary to a matching hypothesis), discipline, assessors chosen by the researchers themselves compared to those chosen by the funding agency, and country of the assessor. Given the large, diverse sample, the powerful statistical analyses, and support for generalizability, these results – coupled with findings from previous research – offer strong support for the null hypothesis model of no gender differences in peer reviews of grant proposals.

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The present investigation is a methodological-substantive synergy (Marsh & Hau, 2007) in which we apply a series of increasingly sophisticated analyses to address a substantively important issue with significant policy implications.

Substantively we address the issue of gender differences in the peer reviews of grant proposals. Peer review is a gatekeeper of what is valued in academia, and is widely used to evaluate grant proposals, journal submissions, job applications, promotions, monographs, textbooks, PhD theses, PhD and post-doctoral applications, and a variety of other academic products (Bornmann & Daniel, 2005; Breckler, 2007; Chubin, 1994; Cicchetti, 1991; Jayasinghe, Marsh, & Bond, 2001; Jayasinghe, Marsh, & Bond, 2003; Jayasinghe, Marsh, & Bond, 2006; Marsh & Ball, 1991). Nevertheless, peer review is widely criticized in terms of potential biases (e.g., Bowen, Perloff, & Jacoby, 1972; Chubin, 1994; Cicchetti & Eron, 1979; Crane, 1967; Gordon, 1980; Mahoney, 1978, 1985, 1987; Merton, 1973; Peters & Ceci, 1982; Wennerås & Wold, 1997; Wennerås & Wold, 2000; Zuckerman, 1970; Zuckerman & Merton, 1971)—particularly in relation to gender as shown by highly cited articles (Wennerås & Wold, 1997; Wennerås & Wold, 2000; also see Cole, 1979; Fox, 1991; Johansson, Risberg,

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Hamberg, & Westman, 2002; National Science Foundation, 2003; Sandström & Hällsten, 2008) and a recent meta-analysis (Bornmann, 2007; Bornmann, Mutz, & Daniel, 2007).

Methodologically we apply a construct validity approach to address critical issues in which we compare and contrast results based on different methodological designs and analyses. More specifically, we demonstrate multilevel cross-classified models that are more appropriate and have broad applicability for applied psychological research. We contend that the richness of advanced quantitative techniques are most appropriately demonstrated when the data and issues are realistically complex—not in isolation based on simulated data, "textbook" examples that are over-simplified, or applications taken out of context of relevant substantive and theoretical issues that drive them. Complex problems typically require appropriately complex analyses and new methodological insights come from addressing realistically complex problems that come from applied research.

1. Substantive issue: gender differences in peer review

1.1. The gender differences and gender similarities: the null hypothesis model

In psychological and social science research – and in applied public policy more generally – there is a pre-occupation with gender differences (Hyde, 2005; also see Marsh, Martin, & Cheng, 2008). Indeed, it is fair to say that gender is one of the most frequently considered variables in applied research. Gender difference research is fueled, at least in part, by a null hypothesis testing perspective. Given a sufficiently large sample size, there will usually be statistically significant gender differences for most outcomes. It is very difficult to prove the null hypothesis as it will typically be rejected, given sufficient power. However, this focus on gender *differences* tends to ignore the strong support for gender *similarities* and the typically small sizes of gender differences actually found. This situation is exacerbated in meta-analyses of gender differences, where there is typically so much power that even tiny gender differences are significant from a purely statistical perspective. As emphasized by Hyde (2005), in her meta-review of meta-analyses of gender differences across many constructs, males and females are more similar than different on most variables. She argued that over-inflated interpretations of small gender differences could result in much harm and lead to counter-productive policies.

Extending Hyde's argument, we consider a "null hypothesis testing fallacy" in which "statistically significant" results are typically interpreted as potentially significant from a substantive perspective, while statistically non-significant results are seen as not being significant from both a statistical and substantive perspective. In contrast, we argue that although it is difficult to establish strong support for the null hypothesis from a mechanistic statistical perspective, it might be highly important to evaluate critically support for a null hypothesis model from a substantive perspective. Support for a statistical model that hypothesizes a null effect of gender typically requires a much stronger study (in terms of statistical power, rigorous design, tests from alternative perspectives, and evaluation of interpretations in terms of their construct validity) than one that merely finds that there is a significant difference.

Indeed, the whole notion of null hypothesis testing has come increasingly under attack as illogical, incoherent, and counter-productive (see Rodgers, 2010), leading some methodologists to recommend that it be abandoned altogether (Schmidt, 1996; also see Harlow, Mulaik, & Steiger, 1997). For example, Gigerenzer (1993, p. 314) argued that current null hypothesis testing practice is "*an incoherent mismash of some of Fisher's ideas on the one hand, and some of the ideas of Neyman and E.S. Pearson on the other hand*." In response to concerns to null hypothesis testing, Rodgers (2010) argued that there is a quiet revolution in the epistemological basis of applied statistics moving away from a mechanistic null hypothesis testing to evaluating statistical and scientific models. In the modeling perspective, the role of the null hypothesis is completely turned around such that instead of seeking to reject the null hypothesis based on mechanistic statistical rules in favor of an unspecified alternative hypothesis, the statistical modeler seeks to retain the hypothesized model as a reasonable approximation of reality based on its ability to fit the data, evaluation of parameter estimates, and an ultimately subjective evaluation of the results (Marsh, Balla, & McDonald, 1988; McDonald, 1997). From this perspective, we refer to the null hypothesis model to emphasize that we are concerned with the viability of model that posits no gender differences. More specifically, in our substantive-methodological synergy, we take on the daunting task of trying to support the null hypothesis model of no gender differences in peer reviews of grant proposals—in relation to the original Bornmann et al. meta-analysis and particularly in relation to the large-scale primary study that is the basis of the present investigation. Consistent with this modeling perspective, we take a construct validity perspective in which we seek to evaluate not only support for the null hypothesis model, but also the robustness and generalizability of interpretations based upon it. We begin with a critical review of the Bornmann et al. meta-analysis.

The Bornmann et al. meta-analysis concluded that there is "evidence of robust gender differences in grant award procedures" (Bornmann et al., 2007, p. 226). Focusing on the important implications of their meta-analysis in a commentary published in Nature, Bornmann (2007, p. 566) concluded that "when it comes to applying for grants, women seem to be at a disadvantage—they are potentially less likely to succeed than their male counterparts" (Bornmann, 2007, p. 566). Given the high profile of these publications, the apparently clear results and unambiguous interpretations, and the strength of the meta-analysis approach that was the basis of their results, the Bornmann et al. conclusions seem to set the standard for what is known in this area. However, a more careful evaluation of specific methodological and statistical issues – including limitations in meta-analysis in general and the Bornmann et al. study in particular – suggests that these conclusions are premature.

1.2. Meta-analysis of gender differences in peer review outcomes

The controversy surrounding potential gender biases in the peer review process – and a lack of conclusive results – led Bornmann et al. (2007) and Bornmann (2007) to pursue a meta-analysis. Based on 66 effect sizes from a total of 353,725 proposals, they found a statistically significant but small effect in favor of men (an effective odds-ratio of 1.07, an effect size of .04 in the traditional 'd' effect size metric). Bornmann et al. argued that their statistically significant effect of gender "can be interpreted as a *robust* average gender effect in grant peer review" (p. 234, italics in original quote). Although it is difficult to assign labels such as small, medium, and large to *ES*s that generalize across studies, the traditional interpretation is that *ES*s of less than .20 are "small" (e.g., Cooper & Hedges, 1994; Lipsey & Wilson, 2001). We do not necessarily endorse such generalized interpretations of effects sizes as an $ES = .2$ can be important in some contexts, but merely note that they are widely used. Nevertheless, in relation to such standards, an *ES* of .04 (approximately 0.2% of the variance explained) must be considered as smaller than small (i.e., very small or tiny). Highlighting this small effect, only one of the 66 gender difference *ES*s in the original Bornmann, Daniel and Mutz meta-analysis was significant based on the 95% confidence intervals in their multilevel analysis (see their Fig. 1).

We also have an alternative interpretation of the term *robust* that was highlighted by Bornmann et al. (2007). Although they used the term to indicate statistical significance, here we use the term to indicate that the effect has broad generalizability across many situations and circumstances rather than mere statistical significance. Hence, it is possible to have a robust non-effect if support for the null hypothesis model has good generalizability or a statistically significant effect that is not robust because it has limited generalizability. This issue is particularly important because Bornmann et al. (2007) found systematic variation in the effect sizes (study-to-study variation) beyond what could be explained by random sampling error, suggesting that their results were not generalizable across even their sample of studies. When there is systematic variation in mean effect sizes, it is essential for meta-analysts to determine how much of the between-study variance can be explained in terms of available study characteristics. In their analysis, important characteristics were not included as explanatory variables, e.g., type of application (grant proposals vs. pre-doctoral and post-doctoral fellowships), discipline area (covering the entire range of social sciences, humanities, bio-medical and physical sciences), country, and year of publication. Recognizing the relevance of this concern, Bornmann et al. (2007, p. 237) noted: "Unfortunately, the inclusion of one or more of these characteristics into the calculation of the meta-analysis resulted in models that did not converge in the estimation process. This finding indicated that the model estimation became too complex by considering specific interaction effects or the included characteristics had no influence on the outcome, respectively."

In order to address this issue, Marsh, Bornmann, Mutz, Daniel, & O'Mara (2009; also see Marsh & Bornmann, 2009) reanalysed and extended the Bornmann et al. (2007) meta-analysis. Consistent with the original results, they found systematic between-study variation. However, particularly relevant to the present investigation, there were no statistically significant gender differences at all for the 40 (of 66) effect sizes associated with grant proposals that are the focus of the present investigation (the remaining 26 effect sizes were for pre- and post-doctoral fellowship applications where there were small gender differences in favor of men). Marsh et al. concluded that the results provided evidence of a very small effect of gender that disappeared completely when the analyses were limited to peer reviews of grant proposals. These non-effects of gender for grant proposals were consistent across studies and did not vary as a function of country, discipline, or year of publication. Therefore, the non-effects of gender were robust. It is also noteworthy that this reanalysis and extension of Bornmann et al. (2007) meta-analysis was co-authored with all the authors of the original meta-analysis, emphasizing the agreement between the two groups in terms of findings and interpretations of their reanalysis. It also provides a model of a constructive collaboration process as an alternative to the potentially divisive series of responses and counter-responses that is still the norm.

1.3. A matching hypothesis: interactions between assessor gender and researcher gender

Because of the nature of the data available to them in their meta-analysis, Bornmann et al. (2007) were not able to consider characteristics associated with individual assessors. This is an inherent limitation of meta-analyses in general, as meta-analysts typically do not have access to raw data from the studies included in the meta-analysis. Making a similar point, Cooper and Patall (2009) argued that if meta-analysts had access to raw data at the level of the individual participants, analyses would be stronger in that they could pursue tests of the generalizability over individual characteristics and relevant interactions that are not possible with aggregated data. This is an important limitation in this in the evaluation of gender differences in peer reviews in that gender effects, to the extent that they exist, must necessarily be based on responses by the assessors. In this respect, evaluating relations between gender of the applicant and assessor characteristics is important. For example, given the focus on effects of researcher gender, a particularly important assessor characteristic is assessor gender. Hence, it is important to extend consideration of gender biases in peer review to include assessor gender as well as applicant gender. The term "matching hypothesis bias" used here (see discussion by Jayasinghe, 2003; also see Bornmann et al., 2007) follows from earlier discussion by Marsh (1987; also see Marsh, 2007) who applied the term in related research on students' evaluations of teaching effectiveness; whether students give higher ratings to same-gender teachers than to opposite-gender teachers. Support for such a matching hypothesis might imply a complex form of gender bias in the teacher evaluations, but there was little or no support for such an interaction (also see Feldman, 1992, 1993).

The basis of the matching hypothesis could also be traced to previous research on demographic similarity, feature matching, or similarity attraction (e.g., Berscheid, Dion, Walster, & Walster, 1971; Tversky, 1977). However, it is important to distinguish our use of the term from other applications where the term is typically used to describe more general aptitude-treatment interactions such that an appropriate matching of individual participant characteristics and those of the situation result in favorable outcomes (e.g., workplace stress and support structures; Terry, Nielsen, & Perchard, 1993) or the finding that people tend to choose others who are similar to themselves (e.g., Berscheid et al., 1971; Kalmijn, 1994). In particular, according to the matching hypothesis bias as used here, the matching results in an actual bias. For example, according to the matching hypothesis, female (male) assessors give more favorable ratings to proposals by female (male) applicants than proposals by male (female) applicants.

2. A methodological-substantive synergy

The focus of the present investigation is a methodological-substantive synergy in which evolving methodological and statistical approaches are applied to substantive issues with important practical implications in applied research—gender differences in peer reviews of grant applications. Consistent with this synergistic approach we carefully distinguish between the substantive issue (gender differences in peer reviews of grant proposals and reconsidering the appropriateness of conclusions by Bornmann et al., 2007,), quantitative approaches (e.g., application of multilevel, cross-classification approaches that facilitate more detailed tests), and methodological design issues (e.g., quasi-experimental comparisons to better evaluate gender differences). Although our central substantive issue is gender differences in peer reviews of grant proposals, methodological and analytic strategies demonstrated here have broad generalizability to the study of rater bias and applied psychological work more generally.

Our investigation is a large-scale, comprehensive primary study in which we evaluate gender differences based on data from the Australian Research Council (10,023 reviews by 6233 external assessors from all over the world of 2331 proposals from all disciplines). The comprehensiveness of this data allows us to evaluate the generalizability of gender differences in relation to individual characteristics of the applicants (e.g., gender, discipline) and assessors (e.g., nationality, gender, type: nominated by applicants or by the funding body) and their interactions in ways that have not been considered previously. Methodologically, we evaluate gender differences in diverse ways not considered in previous peer review research (within-person and within-proposal comparisons, the matching hypothesis, comparisons of author-nominated and panel nominated reviewers), emphasizing that these quasi-experimental strategies have broad generalizability to the study of rater bias and applied research more generally. Importantly, such comparisons typically are not possible with meta-analysis data like those considered by Bornmann et al. (2007), because meta-analysts do not have access to raw data that are required to make such comparisons. We argue that meta-analyses based on secondary data (like the Bornmann et al., 2007, study) and large-scale comprehensive primary studies – like the present investigation – each have complementary strengths and limitations that need to be juxtaposed in order to reach appropriate conclusions, as illustrated here.

Statistically, we demonstrate the application of multilevel, cross-classified models that are more appropriate than the single-level analysis typically used in peer review research. Although such statistical models are not new, even to the area of peer review research (e.g., Bornmann & Daniel, 2005; Jayasinghe et al., 2001; Jayasinghe et al., 2003; also see reviews by Marsh et al., 2009; Marsh, Jayasinghe, & Bond, 2008), they are particularly important in this area of research. Peer review data are inherently multilevel (assessors nested under proposals, proposals nested under discipline groupings) and typically cross-classified (the same reviewer sometimes evaluates more than one proposal). The application of multilevel, cross-classification analyses also facilitates new ways of looking at the data and new tests of substantive issues. The multilevel variance component model with no predictor variables provides an estimate of the variance at each level of the design, including an index of interrater agreement (agreement among different reviewers of the same proposal). It is then possible to add predictor variables representing the individual assessor level (e.g., assessor gender), the proposal level (e.g., applicant gender), and cross-level interactions (e.g., assessor gender × applicant gender), and to test the extent to which each of these effects vary as a function of discipline grouping.

In the present investigation, we also tackle a methodological issue that we refer to as the null hypothesis fallacy. More specifically, we argue that strong support for the acceptance of the null hypothesis model is of practical significance even though such support is more difficult to defend than mere rejection of the null hypothesis. We illustrate the rationale for this perspective by testing the null hypothesis model positing no gender effects in peer reviews—not as a "strawperson" hypothesis designed to be rejected, but as a substantively important model that we seek to support. In our substantive review of research leading to the present investigation, we began with a discussion of the distinction between gender differences and gender similarities, extended this to incorporate the null hypothesis model, and then applied these issues to a critical reanalysis and reevaluation of Bornmann et al. (2007) meta-analysis by Marsh et al. (2009) where results of the present investigation were summarized briefly (also see Jayasinghe, 2003; Marsh, Jayasinghe, et al., 2008) and juxtaposed with the meta-analysis results by Marsh, Bornman et al. In some cases, multilevel analyses merely provide more appropriate tests of essentially single-level hypotheses (e.g., provide more appropriate error terms). However, in many applications, addressing issues from a multilevel perspective leads to substantively relevant new questions that might not be evident from a single-level perspective (e.g., cross-level interactions and tests of generalizability over different units of analysis).

3. Method

3.1. Data set and variables

Data considered here were primarily based on all externally reviewed proposals from a single round of grant applications; 2331 proposals rated by 6233 external assessors (not including the 653 proposals culled in the preliminary round for which there were no external reviews). The external assessors provided a total of 10,023 reviews, an average of 4.3 ratings per proposal. The primary dependent variable in the analyses presented here is the (individual) *assessor rating* for each proposal. This is a weighted average of the project rating (overall quality of proposal, weighted .6) and the researcher rating (overall quality of the research team, weighted .4) by each assessor—the weighting used by the ARC panel to rank grant applications for purposes of funding. These assessor ratings vary at the level of the individual assessors within each proposal. However, in some supplemental analyses we also consider summary scores at the proposal level. The final (aggregate) assessor rating is the average rating across all the external assessors of a given proposal. The final panel rating is the final assessment given by the ARC panel that is based substantially on the ratings by the external assessors (but also based on panel members own reading of the proposal and rejoinders to the assessors by the applicant); final assessor ratings correlate .95 with ARC panel ratings). The final panel ratings are then the basis of the final (dichotomous) success, a simple fund/no-fund dichotomous outcome (success rates were about 21%).

The main independent variables were researcher gender, assessor gender, and their interaction. The gender of the first-named researcher was available for 2981 (8 missing) proposals. Only 15.5% of first researchers were females. Similarly, 15.3% of the total number of researchers were women. Assessor gender was available for 4542 (73%) of 6233 assessors, of which only 430 (9.5%) of the assessors were females. Female assessors reviewed 9.8% of the proposals.

Gender is the primary independent variable considered in the present investigation. We focus particularly on the gender of the first named researcher. However, the gender ratios were similar for first named researchers (15.3% females) and all named researchers (15.5%). Whereas we had nearly complete data (99.8%) on the gender of applicants, there was considerable missing data (27%) for assessor gender. In order to make use of all the available data, we coded assessor gender into three categories (1 = male, 2 = female, and 3 = missing). We then constructed two contrast variables: male vs. female (coded male = −1, missing = 0, female = +1) and missing vs. male + female (coded missing = 2, male = −1, female = −1). Two product terms were created to test interactions between researcher gender and assessor gender that were used to test the matching hypothesis.

There are a total of 144 fields of study that are grouped into nine disciplines where decisions about funding are made. Here, the nine discipline panels were used to construct eight dummy variables—used to test the fixed effects of discipline. Panel 4 (Engineering 1) was chosen as the "left out" reference category as it received the lowest mean assessor ratings. Hence, tests of the other panels were in relation to this reference category. Assessor's country was classified into four regions: Australia (56.6%), North America (19.6%), Europe (18.7%), and "other" (Asia, Africa, South America and New Zealand, 5.1%). For present purposes, missing values for country (2.1%) were classified as "other" allowing us to make use of all the data. Assessor type was classified as applicant nominated assessors (ANAs) or panel nominated assessors (PNAs). All research teams were given the option of nominating up to two assessors, and the ARC typically endeavoured to include at least one ANA. Most proposals (75.7%) had one ANA, whereas some had none (18.9%; typically because the authors of the proposal did not nominate an assessor or the assessor did not respond when asked by the ARC) or more than one (5.4%). There was no missing data for assessor type. For panel, country, and assessor type, cross-product terms were constructed to test interactions between these variables with the applicant gender, assessor gender, and their interaction.

3.2. Statistical analysis: multilevel models

Most forms of peer review ratings have a hierarchical or multilevel structure (e.g., referees for academic journals are nested within manuscripts, assessors of grant proposals are nested within grant proposals, etc.). The dependent variables can be either categorical (e.g., accept, reject, resubmit, for journal articles; fund, no fund for grant proposals) or continuous (e.g., assessor ratings for the quality of grant proposals). Single level models are typically inappropriate if the data have a hierarchical or multilevel structure (Bornmann & Daniel, 2007; Bornmann et al., 2007; Goldstein, 2003; Jayasinghe et al., 2003; Raudenbush & Bryk, 2002).

There are important advantages to the use of multilevel analyses. In particular, it is not necessary to have the same number of lower level units within each higher-level unit (i.e., for the data to be balanced; Goldstein, 2003). For example, the number of reviewers who actually evaluate an outcome (proposal, manuscript, job application, etc.) typically varies from case to case. Because peer review studies typically have unbalanced data, the ability to handle unbalanced data is an important advantage of multilevel modeling. This is in contrast to traditional repeated measures multivariate analysis of variance, which requires balanced data. Importantly, multilevel analyses facilitate the incorporation of data from different levels of analysis into an integrated analytic framework. This is critically important for peer-review research that seeks to evaluate the characteristics of individual assessors nested within each proposal (e.g., assessor gender), characteristics of the proposal and the researchers associated with each proposal (e.g., researcher gender), generalizability across higher level units (e.g., discipline area), and cross-level interactions at different units of analysis. Furthermore, even if interest is focused on only a single level of analysis (e.g., proposal level), the standard errors based on single-level analyses are typically biased, inflating the type 1 error rate (Goldstein, 2003).

In traditional multilevel models, units have a purely hierarchical or nested structure in which units at a lower level (e.g., students) occur at only one level of a higher order unit (e.g., schools). In cross-classified analyses, however, a unit may be classified along more than one dimension (Browne, 2005; Goldstein, 2003; Jayasinghe et al., 2003; Raudenbush & Bryk, 2002; Snijders & Bosker, 1999). Failure to account for such cross-classification can substantially bias estimates of variance explained by the different levels. In the present study, for example, proposals are nested under field of study (each proposal is submitted to one field of study). However, some assessors typically evaluate more than one proposal (so that assessor is not strictly nested under proposal), but each assessor typically evaluates no more than a few of the total number of proposals (so assessor is not crossed with proposal). Hence, peer review data typically have a cross-classified structure. Although cross-classified analyses have not been handled easily by multilevel statistical packages until recently, cross-classified analyses are facilitated by recent developments in multilevel models. These multilevel cross-classification models provide a much stronger statistical basis for testing substantive research hypotheses and questions raised in the present investigation than have been considered previously in this area of research.

For purposes of the present investigation, the data were conceptualized as a three-level multilevel model (Level 1 = assessor; Level 2 = proposal; Level 3 = discipline/field of study) with cross-classification of assessor and proposal (in that some assessors evaluated more than one proposal). MLM analyses were conducted with MLwiN version 2.10, using Markov chain Monte Carlo (MCMC) estimation for cross-classified models (Browne, 2005; also see Rasbash, Steele, Browne, & Prosser, 2005). In preliminary analyses, a baseline variance components model (Rasbash et al., 2005) or intercept-only model (Hox, 2002) was used to evaluate how much variation in the peer reviews could be attributed to each level. Following variance components models, the major focus of analyses was on a set of MLM path models to test the effects of researcher and assessor gender, panel, country of the assessor, type of assessor (applicant or assessor nominated) and various interactions on peer reviews.

Several data transformations were used to facilitate interpretations and infer interaction effects. The response variables project and researcher ratings have skewed distributions with more than 40% of ratings being 87 or above out of 100. Hence, project and researcher ratings are transformed to normal scores and used as the dependent variables (Goldstein, 1995, 2003; Goldstein et al., 1998). To facilitate interpretations in multilevel models, all independent and dependent variables were then standardized. All interaction terms were based on the product of these standardized variables.

4. Results and discussion

The overarching aim of the present investigation is to determine whether the gender of applicants has an effect on the outcomes in the ARC peer review process. There are two quite different perspectives to this issue that provide very different answers. Because only 15.3% of the researchers were females, females are substantially under-represented among those researchers who apply for ARC grants—relative to the number of female academics who are eligible to apply. Although clearly worrisome, the focus of our analysis is on the effects of gender on the proposals that were submitted. We present the results in two stages. First, we begin with preliminary results based on single-level analyses and associated descriptions of the data. Because these preliminary analyses are at the level of the proposal, single level models might be justified (but still ignore cluster effects associated with field of study). However, for the primary analyses at the level of assessor within proposal, we turn to multilevel cross-classified analyses of the data. In this respect, the preliminary (single-level) analyses provide an initial overview of results that are pursued in subsequent (multi-level) analyses.

4.1. Preliminary results: single-level analyses

4.1.1. Researcher gender effects on proposal success

We begin our analyses with a summary of researcher gender effects on the dichotomous success (fund/no fund outcome) for each proposal. Single-level analyses might be justified in that this outcome is at the proposal level. The evaluation of dichotomous fund/no fund variable is also different from those based on the panel and assessor ratings in that it includes the 653 proposals that were culled in the preliminary round; as it was appropriate to conclude that these had not been funded. Because the culled proposals were not sent to external reviewers or considered further, there were no panel or external assessor ratings for these proposals. These analyses of the effects of research gender are also complicated by the fact that there could be up to four named researchers for each proposal. Whereas all proposals had at least a first-named researcher, decreasing numbers had 2 or 3 named researchers and only a very few had more than 3 named researchers. For purposes of these initial analyses, we conducted separate analyses on results for the first, second, and third named researchers, but place particular emphasis on those based on the first-named researcher.

Only 15.3% of the ARC applications were by women. However, the percentage of successful applications by female researchers (15.2%) is almost exactly proportional to their representation (see Table 1). When gender of only the first-named researcher was considered, the success rate of both males and females was 21.3%. Logistic regression analyses were conducted using the dichotomous dependent variable success (1 = success, 0 = no success) and a dummy variable for researcher gender as the independent variable. A separate logistic model was conducted for the first, second, third and fourth researchers. The effect of researcher gender on success was statistically nonsignificiant in all four models, indicating that the researcher's gender had no effect on the success of the proposal.

Table 1
Gender of researchers across all disciplines (including initial culls).

Gender	1st named researcher	2nd named researcher	3rd named researcher	4th named researcher	All researchers
Males					
No. of applicants	2520 (84.5%)	1289 (85.4%)	414 (84.5%)	33 (75.0%)	4256 (84.7)
No. of successes	537 (84.6%)	262 (83.7%)	93 (88.7%)	8 (100%)	900 (84.8%)
Success rate	21.3%	20.3%	22.5%	24.2%	21.1%
Females					
No. of applicants	461 (15.5%)	220 (14.6%)	76 (15.5%)	11 (25.0%)	768 (15.3%)
No. of successes	98 (15.4%)	51 (16.3%)	12 (11.3%)	0 (0.0%)	161 (15.2%)
Success rate	21.3%	23.2%	21.3%	0.0%	20.1%
Total					
No. of applicants	2981 (100%)	1509 (100%)	490 (100%)	44 (100%)	5024 (100%)
No. of successes	635 (100%)	313 (100%)	105 (100%)	8 (100%)	1061 (100%)
Success rate	21.3%	20.7%	21.4%	18.2%	21.1%
Wald statistic[a]	0.0008 (NS)	0.9198 (NS)	1.6765 (NS)	0.0335 (NS)	

Notes: 1st–4th named researcher refers to the first through fourth researcher listed on the research proposal. ALL, all proposals; SUC, number of successful proposals. Separate values are provided for the first, second, third and fourth named researcher (although 2981 proposals were authored by the first named researcher). The figures in parentheses are the success rates (% of proposals that are successful). Z values provide a test of whether the success rates for a given group differ significantly (Z values > 1.96) from that of the total sample.

[a] A logistic regression related researcher gender to success. Four separate analyses were done for the four named researches. The Wald statistic was not statistically significant (Wald < 3.85, corresponding to $p < .05$) for any of the four analyses.

4.1.2. Researcher gender effects on final panel and aggregated assessor ratings

Here we evaluate the effect of researcher gender on the (proposal level) aggregated assessor rating and the final panel rating. For these analyses, the culled ratings are excluded (as there were no external assessor or panel ratings for these excluded proposals). Linear regression analyses were carried out with the final panel and aggregated assessor ratings as the dependent variable and the gender of researchers as the independent variable. The linear regression analyses showed that the gender of first, second, and third named researcher had no significant effect on the panel-committee ratings (all ps > .10), as was the case for the final (aggregated) assessor ratings. These results and findings parallel those for the logistic regression analyses of the dichotomous success variable. Because both sets of analysis showed that there was no difference in the results of the gender for the first-, second-, or third-named researcher (and many proposals had only one named researcher), the focus of subsequent analyses is on the gender of the first-named researcher.

Next, we considered the extent to which the lack of gender differences in the final assessor and final ARC panel ratings generalized across the nine ARC discipline panels. We conducted a multivariate analysis of variance (ANOVA) using the gender of the first-named researcher and the nine panels as (between group) independent variables (see Fig. 1). The main effect of researcher gender was not significant for either the final ARC panel or the aggregated assessor ratings (ps > .20), but there were significant differences between panels for both these proposal-level ratings (ps < .001). However, of particular importance, the interaction between researcher gender and the ARC discipline panel was not significant for either panel ($F[82,302] = 1.04$, $p = 0.4$) or aggregated assessor ($F[82,302] = .81$, $p = 0.59$) ratings. Therefore, the (non)effect of gender did not interact significantly with panel, demonstrating the lack of gender effect generalized well over the nine social science, humanities, and science disciplines. Hence, for this large multidisciplinary archive of peer reviews, there was no evidence of a gender effects in the reviews of applications by men and women researchers and support for the null hypothesis model of no gender differences.

4.1.3. Gender of the assessor and the matching hypothesis: a within-proposal perspective

Next, we evaluate the effect of the gender of the assessor, recalling that the gender was supplied by only 73% of the assessors (4542 of 6233). Surprisingly, only 430 (9%) of the assessors were female, although this varied substantially with the panel. For example, the largest percentages of female assessors were from the social sciences (34%) and humanities (23%) panels. Because of the complicated hierarchical structure of these data and the fact that many proposals had no female assessors, we begin with a preliminary analysis using a simple within-proposal perspective.

For the within-proposal analyses, we only considered the 584 proposals that had at least one male and at least one female assessor. For each proposal, we compared the mean of ratings by male assessors and the mean of ratings by female assessors for a given proposal (i.e., a within-proposal perspective). Simple paired t-tests based on this within-proposal comparison indicated that there were no significant differences between ratings of male and female external assessors ($p > .10$). An advantage of this within-proposal comparison is the ability to control all characteristics associated with the proposal and applicants, including the quality of the proposal, by holding the proposal constant. Although insightful, the limitation of this within-proposal perspective is the inability of the single-level analysis to handle unbalanced data and the fact that so few proposals had female assessors so that on only a small portion of the proposals were considered.

Because we know the gender of both the researchers and the external assessors for most proposals, we had a unique opportunity to determine the interaction between these two variables. More specifically, we tested a “matching hypothesis”

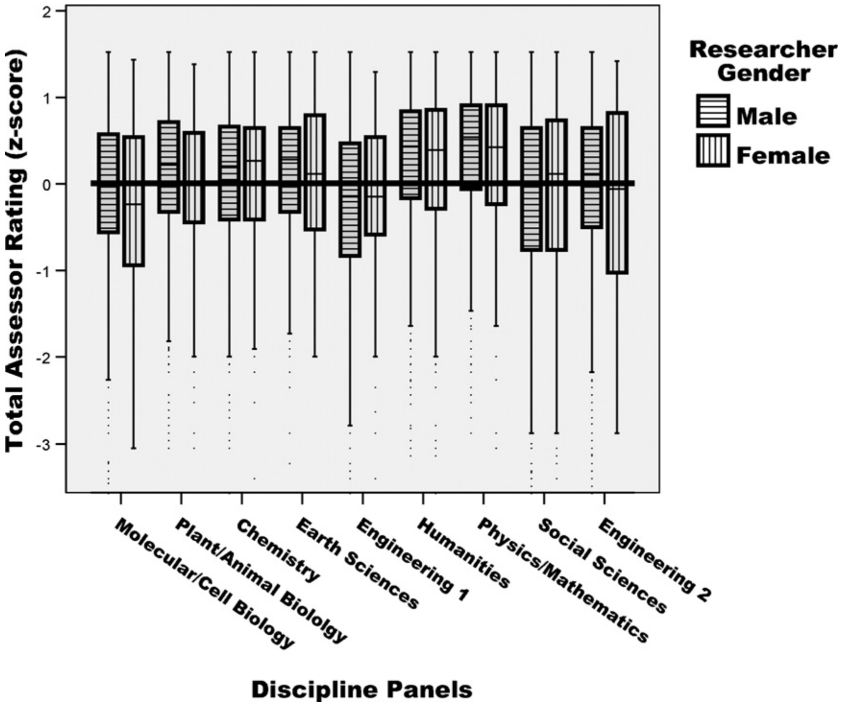

Fig. 1. Total assessor rating (in *z*-score, standard deviation units) as a function discipline panel and gender of the first-named research. The results are depicted as box plots in which the boxes represent the 25th and 75th percentiles (and the median as the dark line within the box). The lines above and below the boxes represent the 90th and 10th percentiles respectively.

based on the hypothesis that female external assessors give higher ratings to female researchers and that male external assessors give higher ratings to male researchers. Again, we begin with preliminary analyses based on a within-proposal perspective, considering only the mean ratings by male and female assessors for each of the proposals with at least one male assessor and at least one female assessor. We then conducted a repeated measure ANOVA in which we tested the main effect of the gender of the first-named researcher (a between subject effect), the main effect of assessor gender (a within subject effect), and their interaction. For this analysis, none of the main effects nor the interaction effect was statistically significant (all $ps > .20$). Of particular relevance to tests of the matching hypothesis, female assessors did not favor female (or male) researchers and male assessors did not favor male (or female) researchers. Hence, there is no support for a gender-matching hypothesis. Whereas the within-proposal rationale provides a strong basis of comparison by controlling so many potentially confounding variables (by holding the proposal constant), the analyses were based on only small proportion (12%) of the proposals.

4.2. Multilevel gender effects and interactions with researcher and assessor characteristics

We now turn to a set of multilevel models that are the main focus of our analyses based on the separate ratings of each assessor of each proposal. Each model has three levels: the discipline field of study, the proposals, and the assessor (the random effects in Table 2). We begin with a simple model with no predictor (independent variables). This variance component model indicates how much variance there is at each level. The largest variance component (.815) is for differences among assessors of the same proposal. However, there is also substantial variance at the proposal level (.144), indicating systematic variation among proposals based on the aggregated assessor rating. Finally, there is a small but significant variance component (.052) associated with field of study, indicating that there are systematic differences between the 144 fields of study considered in the ARC peer review process (although field of study is considered as a random variable we also consider models in which the nine discipline panels are considered as predictor variables; also see Fig. 1).

4.2.1. Researcher gender effects on proposal success

In the set of analyses based on Model 1, we consider the effects of the gender of the first named researcher, adding the fixed effect of gender to the variance components model already considered. In Model 1A, we evaluate only the effect of researcher gender. The effect (−.012 in Table 2) is close to zero and statistically non-significant in relation to its standard error (.013). There is no significant effect of gender on assessor ratings. Consistent with this lack of effect of researcher gender, the variance components associated with assessors, proposals, and discipline are nearly unchanged (compared to the variance component Model 0).

Table 2
Multilevel models of gender differences in overall ratings: in combination and interaction with other assessor and research characteristics.

Response variable	M0 variance component model	M1A researcher gender female (RF)	M1B researcher gender female (RF)	M1C researcher gender female (RF)	M2A assessor gender female (AG:FvM)	M3A reseacher nominated assessor (RNA)	M4A assessor counrtry
Fixed effects							
Intercept	.042(.026)	.041(.026)	.041(.026)	–.010(.019)	–.011(.020)	–.012(.020)	–.011(.019)
Researcher female (RF)		–.012(.013)	–.012(.014)	–.020(.015)	–.018(.015)	–.020(.015)	–.024(.016)
P0				.027(.028)	.026(.028)	.025(.027)	.026(.027)
P1				.084(.028)	.089(.028)	.082(.028)	.074(.028)
P2				.081(.022)	.076(.022)	.075(.022)	.071(.022)
P3				.075(.018)	.070(.018)	.073(.018)	.063(.018)
P5				.156(022)	.163(022)	.151(022)	.149(022)
P6				.172(.021)	.168(.021)	.159(.021)	.147(.021)
P7				.029(.027)	.037(.027)	.029(.027)	.029(.027)
P8				.066(.024)	.066(.024)	.058(.024)	.048(.024)
RF × P0				–.030(.019)	–.030(.019)	–.029(.019)	–.027(.019)
RF × P1				–.033.019)	–.034(.019)	–.030(.019)	–.032(.019)
RF × P2				–020(.021)	–019(.021)	–013(.021)	–015(.021)
RF × P3				–.015(.019)	–.013(.019)	–.013(.019)	–.017(.019)
RF × P5				–.013(.018)	–.012(.018)	–.011(.018)	–.010(.018)
RF × P6				–.021(.022)	–.020(.022)	–.021(.022)	–.024(.022)
RF × P7				–.004(.019)	–.003(.020)	–.003(.020)	–.001(.020)
RF × P8				–.023(.026)	–.022(.026)	–.023(.026)	–.026(.026)
AG: FvM (female)					–.034(0.12)	–.024(0.12)	–.023(0.12)
AG: missing					–.090(.012)	–.040(.012)	–.028(.012)
RF × AGFvM					–.005(.009)[a]	–.004(.009)	–.001(.009)
RF × AGMiss					.018(.011)[a]	.009(.011)	.014(.011)
Nom						.251(.009)	.239(.009)
RF × NOM						.016(.009)[a]	.011(.009)
AGMvF × NOM						.000(.010)[a]	.000(.010)
RFxAGMvFxNom						–.013(.009)[a]	–.009(.009)
Country-NAmer							.085(.010)
Country Europe							–.015(.010)
Country-other							.018(.010)
RF × NAmerica							.025(.030)[a]
RF × Europe							.015(.011)[a]
RF × other							.023(.011)[a]
Random effects (variance components							
Discipline (Level 3)	.052(.011)	.052(.011)	.054(.011)	.014(.006)	.014(.006)	.014(.006)	.014(.006)
Proposal (Level 2)	.144(.011)	.144(.011)	.143(.011)	.145(.011)	.148(.011)	.163(.011)	.160(.011)
Assessor (Level 1)	.815(.013)	.815(.013)	.815(.013)	.815(.013)	.807(.013)	.733(.012)	.728(.012)
RF Rand at L3			.001(.001)				

Note: Models are labeled for ease of reference in the text. All parameter estimates are completely standardized (i.e., all predictor variables and outcome variables were standardized, $M = 0$, SD = 1) and values in parentheses refer to standard errors of each parameter estimate. In each successive model a new predictor variable was added (along with its interactions with researcher gender). The nine discipline panels are: P0, Molecular/Cell Biology; P1, Plant/Animal Biology; P2, Chemistry; P3, Earth Sciences; P4, Engineering 1 (electrical and computing); P5, Humanities; P6, Physics and Mathematics; P7, Social Sciences; P8, Engineering 2 (civil, mechanical, and chemical).

[a] In supplemental analyses, these terms were allowed to be random (as in Model 1B) at the discipline level and to interact with the set of variables representing panel (as in Model 1C). However, as not of these additional effects were statistically significant (and to conserve space), these additional models are not reported.

A very important feature of our data is that we have results from the entire range of social science, humanities, and science disciplines. These are represented by the 144 discipline fields of study (considered to be a random variable in these models) and the corresponding nine discipline panels (considered to be a fixed effect in these models, represented by a set of eight dichotomous dummy variables). Hence, a critical aspect of the analyses is to determine the extent to which the non-effect of gender generalizes over discipline. We pursue this research question from two different perspectives.

In Model 1B, we allow the fixed effect of researcher gender to be random at the discipline level. This means that the effect of gender was allowed to vary from discipline to discipline. The variance component associated with the effect (RG random at field of study in Table 2) was not statistically significant. Hence, the non-effect of gender did not vary significantly across the 144 fields of study.

In Model 1C we took a different perspective to this same issue, based on the fixed effects associated with the discipline panel. Here we added fixed effects to represent the nine discipline panels and the interaction between research gender and discipline panel. The main effects of the discipline panel were represented by eight dichotomous dummy variables in Table 2. Many of these were statistically significant, indicating that there were differences between panels in terms of overall ratings of the proposals (also see Fig. 1). It is also interesting to note that the variance component associated with field of study dropped substantially (from .054 to .014) with the inclusion of panel as a fixed effect. Although variation among fields of study was still significant, much of this variance – not surprisingly – was explained in terms of differences between panels. However, for purposes of the present investigation, our main concern is the interaction between researcher gender and discipline panel. None of the eight cross-product terms representing this interaction is statistically significant. Hence, the non-effect of researcher gender does not vary across (i.e., interact with) the discipline panels.

In summary, in support of the null hypothesis model of no gender effects, the results of this set of analyses showed that there was no effect of research gender on the ratings of the individual assessors of each proposal. Furthermore, this non-effect of researcher gender generalized well across the 144 fields of study and across the 9 discipline panels.

4.2.2. Gender of the assessor and the matching hypothesis

In Model 2 we extended our analyses by including assessor gender and its interaction with researcher gender. This provides a test of the matching hypothesis positing that female assessors give higher ratings to female researchers than to male researchers, and that male researchers give higher ratings to male researchers than to female researchers. However, this analysis is complicated by the fact that there was missing data on the gender for 27% of the assessors. Although there is a growing sophistication in how to best handle missing data, the most defensible approach for categorical data is to simply include it as a separate category. This strategy allows us to use all of the data (unlike alternatives such as listwise deletion for missing data) and does not make any assumptions about the nature of the missing data (unlike various imputation strategies). Hence, assessor gender is represented by three categories (male, female, missing) that were used to construct two contrast variables: one representing the difference between male and female assessors (AG:MvF in Table 2, coded male = −1, female = 1, missing = 0) and one representing the missing data (AG:Missing in Table 2, coded male = −1, female = −1, missing = 2).

In Model 2A, the main effect of researcher gender and its interaction with discipline panel remained statistically non-significant. There was, however, a small but statistically significant effect of assessor gender. In particular, male assessors gave somewhat higher ratings than did female assessors. This effect was statistically significant (.034 with a standard error of .012 in Table 2). In addition, assessors who had missing values on gender gave slightly lower ratings than did other assessors (−.090, SE = .012). Although these main effects of assessor gender are interesting in their own right, our primary interest is in the interaction of researcher gender with assessor gender that is used to test the matching hypothesis. However, neither of the crossproduct terms representing this interaction was statistically significant. In supplemental (unreported) analyses we evaluated the extent to which this non-significant interaction effect generalized across field of study (as in Model 1B) and discipline panel (as in Model 1C, evaluating the three way interaction between research gender, assessor gender, and panel). Again, however, none of the added effects was statistically significant.

In summary, the results based on Model 2 indicated that there was no significant interaction between researcher gender and assessor gender. Furthermore, these results were consistent across discipline. Although male assessors gave slightly higher ratings than female assessors (.034 SD), this small effect was similar for proposals by male and female researchers. In conclusion, there is absolutely no support for the matching hypothesis and good support for the null hypothesis model of no interaction between assessor gender and researcher gender.

4.2.3. Effect of researcher nominated assessors

In Model 3, we extended our analyses by including the effect of the type of researcher into the analysis. As noted earlier, the ARC previously allowed researchers the option of nominating several assessors to evaluate their proposal and the ARC attempted to include at least one applicant-nominated assessor (ANA) for each proposal as well as other assessors nominated by the ARC panel (PNAs). Previous research (Marsh et al., 2007; also see Marsh, Jayasinghe, et al., 2008) indicated that ANAs were biased and that their ratings systematically detracted from the reliability and validity of the proposal evaluations. Furthermore, when the same assessor evaluated different proposals in the role of an ANA and a PNA, only their ratings as an ANA were biased. This led to the ARC discontinuing the strategy of using ANAs. Although substantively important in its own right, the question addressed here is whether the non-effect of researcher gender and lack of support for the matching hypothesis (researcher gender × assessor gender) generalizes over ratings by ANAs and PNAs. Thus, for example, it is plausible that female assessors specifically nominated by female researchers might give higher ratings to the proposal even if this type of bias did not occur for assessors nominated by the panel (i.e., PNAs)—particularly given that ANA reviews are known to be positively biased.

Extending Model 3, four new effects were added, the main effect of ANA (assessor type) and its interaction with research gender, assessor gender, and their interaction. As shown in previous research, there was a substantial positive effect of ANAs (.251 SDs). Furthermore, consistent with previous interpretations, controlling for this ANA effect decreased the size of the assessor variance component (less error due to disagreement among raters of the same proposal so that the ratings were more reliable) and increased the proposal variance component (better systematic differentiation among proposals so that the proposals were more valid). However, this large effect of ANAs did not interact significantly with researcher gender, assessor gender, or the researcher gender × assessor gender interaction. In supplemental (unreported) analyses we evaluated the extent to which these non-significant interaction effects generalized across field of study and discipline panel as in earlier analyses, but again none of the added effects was statistically significant.

In summary, in support for the null hypothesis model the non-effect for researcher gender was consistent across assessors nominated by the researchers themselves and assessors nominated by the ARC panel. Similarly, the lack of support for the matching hypothesis generalized across ratings by ANAs and PNAs. Furthermore, these non-significant effects were consistent across disciplines. Hence, even though there was good support for a substantial bias associated with ANAs in terms of ratings overall, there was no evidence that there was any gender bias in their ratings.

4.2.4. Effect of country of assessors

In Model 4 we extended our analyses by including the country of the assessor. Whereas a majority of the assessors was from Australia, sizable numbers were from other countries as well. For purposes of the present investigation, we classified assessor country as Australia (the reference category), North America (mostly USA), Europe, and "other" (including the small number of assessor reports for which country was missing; 2.1%). There was a main effect of country represented by the substantially higher ratings by North American assessors. Although substantively interesting in its own right (for further discussion, see Marsh, Jayasinghe, et al., 2008), the main focus of the present investigation was whether this effect of country interacted with the effect of researcher gender. However, none of these three interaction effects was statistically significant. Furthermore, in supplemental (unreported) analyses these non-significant interaction effects generalized across field of study and discipline panel. In summary, the non-significant effect of research gender generalized well over country.

5. Summary and conclusions

In summary, the peer review process is highly valued but widely criticized as the primary basis for evaluating what is good within academic settings. Despite its importance and long history of controversy, there has been surprisingly little empirically rigorous and sound research on posited gender biases. In one of the largest and most ambitious studies to date, we demonstrate strong, robust results in support of a gender similarity hypothesis and the null hypothesis model of no gender differences—and no support at all for systematic effects of researcher gender on peer review outcomes. Importantly, there was extremely good support for the genderalizability of these conclusions in relation to assessor characteristics, nationality, discipline and particularly gender.

5.1. Multilevel cross-classified models, construct validity, and generalizability

Methodologically we extended the multilevel cross-classified approach that is more appropriate for peer review. Although the issue of cross-classification has been recognized for a long time, it is not routinely included in multilevel analyses. However, the argument for its routine application is essentially the same issue as using an independent *t*-test when a paired *t*-test is more appropriate (because of the paired nature of the data—the dependencies). There may be special cases in which the use of the wrong statistical test (independent *t*-test) might be acceptable or at least will lead to the same conclusions as the more appropriate test, but clearly the default should be to use the correct test (paired *t*-test). More generally, we contend that the appropriateness of cross-classification is really a function of the nature of the data—if that data are cross-classified, then a cross-classified model should be used.

Given the inherent complexity of the interpretations of gender difference in peer reviews from a mechanistic null hypothesis perspective, we approached the issue from the perspective of a null hypothesis model based on a construct validity approach in which we used the convergence of different approaches to draw out conclusions and test alternative interpretations. In particular, results from both the "within proposal" perspective that held the proposal constant agreed with our more extensive multilevel analysis. Extending our multilevel approach to include appropriate tests of cross-level interactions, we were able to demonstrate the generalizability of the non-effect of gender across assessor and proposal characteristics.

Pursuing our construct-validity approach in relation to gender differences, we explored two situations in which gender differences might be more likely. First, we extensively evaluated support for the matching hypothesis, namely that assessors would give higher ratings to applicants of the matching gender. However, a variety of different analyses showed no support whatsoever for the matching hypothesis and there was good evidence for the generalizability of this lack of support across discipline and nationality. In this respect, the present investigation extends previous research on the matching hypothesis, particularly as applied to peer review.

Although less widely studied, previous research has identified a bias in the ratings by reviewers nominated by the applicant (ANAs) compared to those nominated by the funding panel (PNAs). While interesting in its own right, our focus is to juxtapose this known bias with potential gender differences. The very strong effect of the ANA-bias (Table 2) demonstrates that there is at least one source of bias in these data and shows that the analyses can identify sources of bias. Furthermore, we reasoned that if ANAs are biased in favor of applicants who nominated them, then their results might show a gender bias as well. However, analyses specifically designed to test this speculation demonstrated that the non-effect of applicant gender and the non-support for the matching hypothesis generalized well across assessors nominated by the funding panel and those nominated by the applicant. Hence, even in this situation where assessors are known to be biased in their ratings, there is no evidence of gender differences.

It is also important to acknowledge potential limitations in our research program, in order to provide appropriate caveats in relation to our interpretations and to provide directions for further research. Because our research is largely correlational, it provides a weak basis for causal inferences. This is particularly the case in evaluating potential biases. Whereas use of multiple, converging approaches may provide reasonable support for the construct validity of interpretations, causal inferences must always be made cautiously. For this reason, we have focused specifically on gender effects, avoiding the interpretation of results as a systematic bias that implies causality. Whereas stronger statistical tools and application of a construct validity approach are useful in countering these problems, interpretations of bias can only be defended in relation to a well-articulated operational definition of bias and construct validity research demonstrating potential sources of bias do not reflect valid sources of variance in relation to multiple validity criteria—as is the case in other applied evaluation research (e.g., Marsh, 2007). Particularly important are our results showing that the lack of gender differences associated with researcher gender generalized across (i.e., did not interact with) the gender of the assessor, the basis by which a particular assessor was selected (researcher nominated or panel nominated), or the discipline area of the proposal (ranging across the social science, humanities, and science disciplines considered here).

5.2. Juxtaposition between large, comprehensive primary studies and meta-analysis

In concluding, it is also relevant to juxtapose our study with the Bornmann et al. (2007) meta-analysis and the reanalysis of this data by Marsh et al. (2009). There are important strengths and weaknesses associated with both meta-analyses and large, comprehensive primary studies like our research reported here. Meta-analysis is an important tool that epitomizes the essence of good science—testing the replicability and generalizability of results, and study characteristics that explain differences when the results are not consistent. Tests of generalizability are difficult to accomplish in even a “gold standard” single primary study in that the main strength of the scientific method is that it controls most sources of variation—typically by holding them constant. Although providing a strong basis for hypothesis testing, a single study is inherently weak in terms of evaluating the generalizability of the results in relation to other variables that are not included or held constant (e.g., sample of participants, testing materials, research design, statistical analysis, nature of the intervention, experimental control). Hence, a comprehensive meta-analysis is valuable in evaluating the consistency of the results across different studies. Meta-analyses, however, are limited in that researchers do not have access to the original (raw) data and can only draw conclusions in relation to variables that are included (and reported) in the original studies. Particularly when there is significant study-to-study variation, it is typically difficult to determine what study characteristics might explain those differences because there is insufficient information about each of the studies. Importantly, meta-analyses are almost entirely limited to looking at study-level characteristics and are rarely able to evaluate characteristics of individual participants (e.g., interaction between researcher gender and assessor gender that is a focus of the present investigation). Indeed, even when potentially important interaction effects at the individual participant level are reported in a few studies, it is likely that many studies do not report the appropriate interactions so that it is difficult to incorporate them into the meta-analysis. Hence, the position taken here is that good meta-analyses and strong primary studies are complementary, and should be considered together in drawing conclusions. Although results of the present investigation seem to contradict interpretations of their meta-analysis by Bornmann et al. (2007), the reanalysis of the results by Marsh, Bornmann, et al. demonstrated that the effect size in the original meta-analysis was tiny (an effect size of .04 in the traditional “d” effect size metric). Importantly, when consideration was limited to only the 40 effects sizes for grant proposals relevant to the present investigation (26 effect sizes were for pre- and post-doctoral fellowship applications), there was no statistically significant effect of gender at all and no significant variation between studies in terms of this non-effect of gender. Furthermore, because of the large sample size, the non-significant effect of gender had a narrow confidence interval—thus providing strong support for a gender-similarity hypothesis and the null hypothesis model of no gender differences. A cautious juxtaposition of the two studies might be that both show that gender differences in peer reviews of grant proposals are extremely small, that the near-zero effect sizes have a very small confidence interval (due to the substantial power in both studies) and generalize over a wide variety of different circumstances and study characteristics.

Given the complementary strengths and weaknesses of meta-analysis juxtaposed with those of our comprehensive primary study, the results are compelling. The meta-analysis results demonstrate the generalizability of the non-effects of gender across a wide range of different studies, different funding agencies, and over time—characteristics not easily considered in primary studies. Our primary study presented here demonstrates that the non-effect of researcher gender generalizes over many characteristics of the assessors (gender, country, type)—features that could not be considered in the meta-analysis. Both studies showed that the non-effect of gender generalizes over discipline and different aspects of country

(country of the funding agency in the meta-analysis, country of the assessor in our primary study). Although it may never be possible to prove the null hypotheses in a purely statistical sense, this juxtaposition of results provides good support for the null hypothesis model. Taken together the results provide very strong support for Hyde's (2005) gender similarity hypothesis – and its generalizability across a wide variety of potential moderating variables – and no support what-so-ever for a gender differences in peer reviews of grant proposals.

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